MACHINE LEARNING · PBL REVIEW · UNITS 1 – 4

EarlyDrop

Student success and dropout-risk analysis. Spotting academic risk before the grades fall.

SYNTHETIC COHORT · n = 400 · CALIBRATED ON 15 REAL RESPONSES

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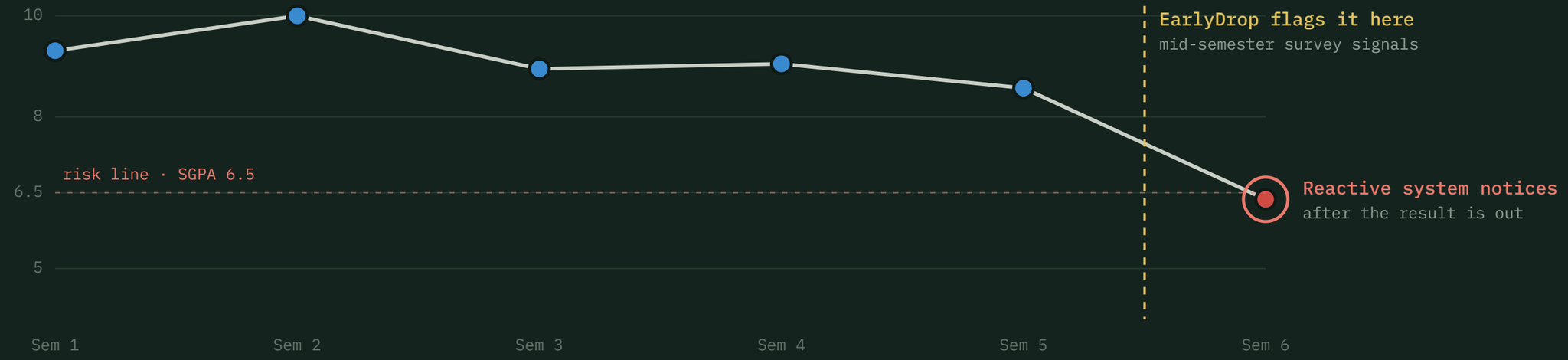
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PROBLEM IDENTIFICATION

Help usually arrives after the grades have already fallen.



QUESTION

Can mid-semester survey signals flag students whose **end-semester SGPA will fall below 6.5**?

SIGNALS

Attendance, internal marks, backlogs, study behaviour, engagement, motivation, support.

USE

An early-warning score that triggers an **offer of help**. Never a label.

DATASET SELECTION

A live Google Form, and an honest problem with it.

15 real responses

CGPA 8.28 – 10 · 0 backlogs in total

SO WE

generate **400 synthetic students** in the form's exact 50-column format, and **calibrate** their answers so high performers match the real ones.

MEAN GAP AFTER CALIBRATION

0.011

Likert points, averaged over 28 items



PRE-PROCESSING

The real export was messy, so the synthetic one is too.

01

406

raw form rows

Google Form export, 50 columns

02

6

duplicate submissions

same answers, seconds apart

03

22

”80.50%” strings

parsed to numbers

04

32

rows from before

9 questions were added mid-survey

05

4

impossible values

78.6 · 6% · 6% · 108

06

292

cells imputed

median / median category

07

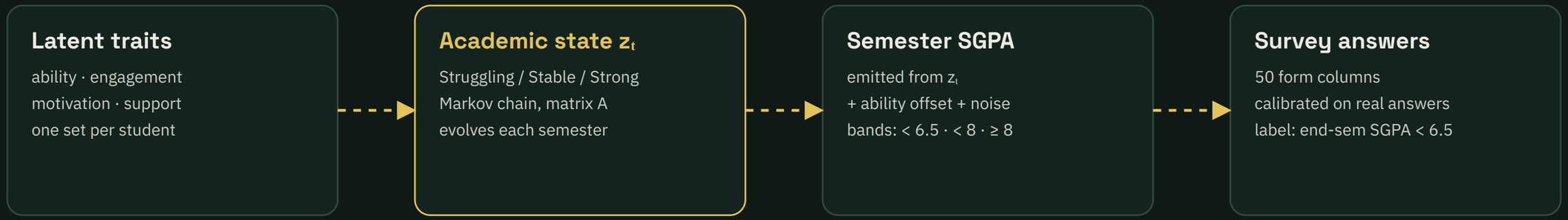
400

clean students

49 modelling columns, 0 missing

HOW THE COHORT IS GENERATED

A hidden process we control, so every method can be checked.



Ground truth we keep, so Units 3–4 can be checked:

- true transition matrix A
- true hidden state per semester
- the 4 latent traits
- calibration targets from the real form

true self-transitions: $A_{11} = 0.78 \cdot A_{22} = 0.78 \cdot A_{33} = 0.82$

PROJECT REVIEW 1 · RECAP

PBL-I

Units 1 & 2

EDA · covariance · correlation

Bayes' rule

Curve fitting

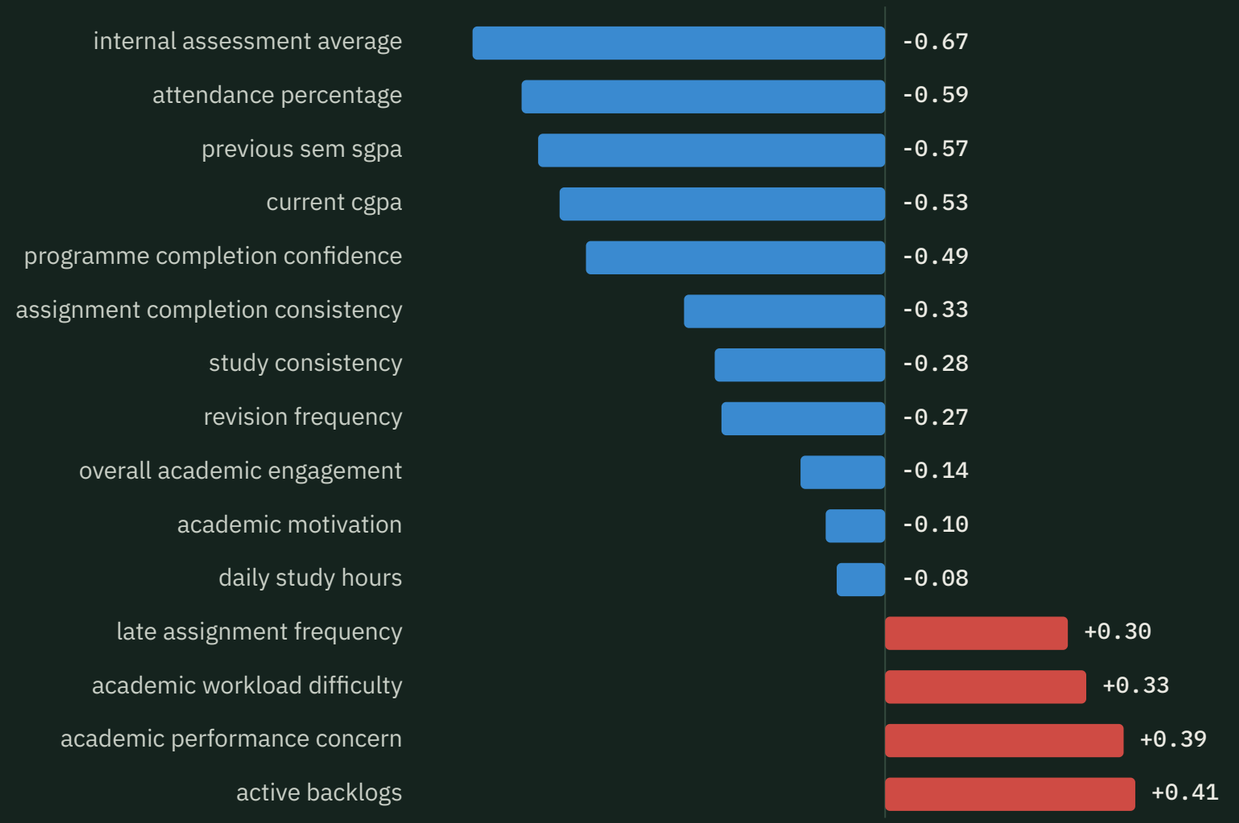
OLS · Ridge · Huber · Bayesian LR

Logistic · Bayesian logistic · LDA · SVM

UNIT 1 · EXPLORATORY DATA ANALYSIS

Which signals move with risk?

PEARSON R WITH AT_RISK (15 SELECTED FEATURES)



111 / 400

students at risk (end-sem SGPA < 6.5)

COVARIANCE → CORRELATION

PAIR	COV	R
attendance percentage ↔ internal assessment average	73.98	0.63
daily study hours ↔ current cgpa	0.41	0.27
academic motivation ↔ overall academic engagement	0.40	0.36
active backlogs ↔ current cgpa	-0.88	-0.61

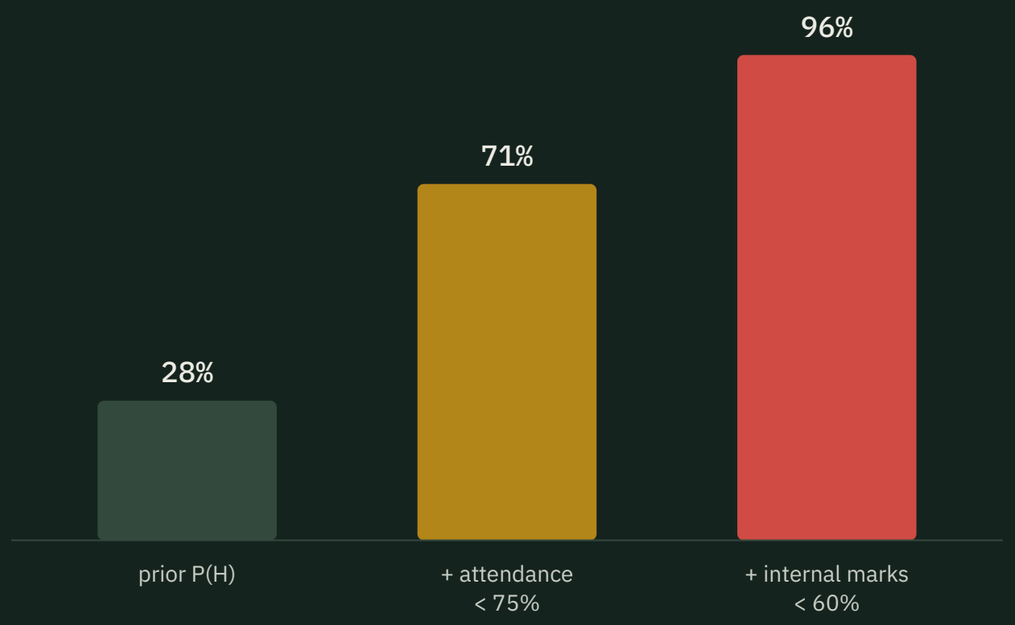
EXPECTATION

$E[\text{backlogs}] = \Sigma\, x \cdot P(x) = \mathbf{0.522}$

UNIT 1 · PROBABILITY & CURVE FITTING

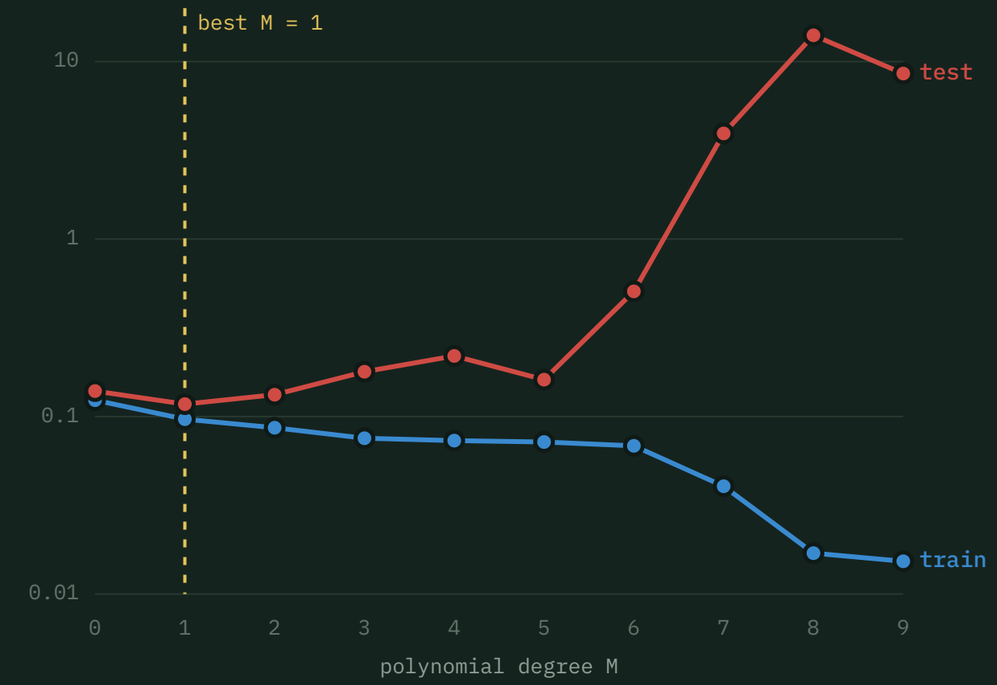
Evidence updates belief. Too much flexibility memorises noise.

BAYES' RULE · P(H | E) = P(E | H) P(H) / P(E)



Exact P(H | both) from the data: **0.963** on 54 students. Naive gap +0.002.

POLYNOMIAL FIT · 12 TRAINING STUDENTS · RMSE (LOG SCALE)



M = 9 test RMSE **8.56** → with ridge (ln λ = −8) **0.208**

UNIT 2 · LINEAR MODELS FOR REGRESSION

Predicting end-semester SGPA, with error bars.

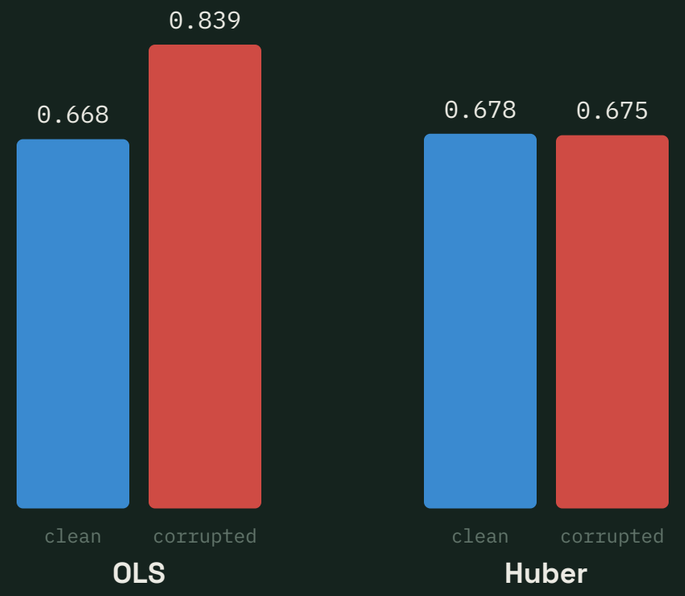
HELD-OUT TEST SET

MODEL	R^2	MAE	RMSE
OLS (normal equations)	0.789	0.534	0.668
Ridge	0.798	0.516	0.654
Huber (robust)	0.783	0.549	0.678
Bayesian Linear Regression	0.789	0.534	0.668

93%
inside the Bayesian 95 % band

16.0
effective parameters γ (evidence)

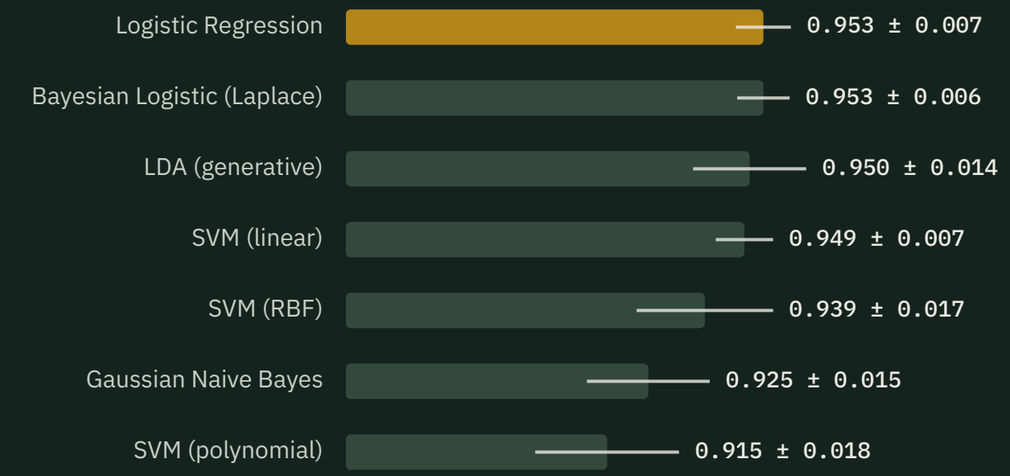
ROBUST REGRESSION · 8 % CORRUPTED TARGETS



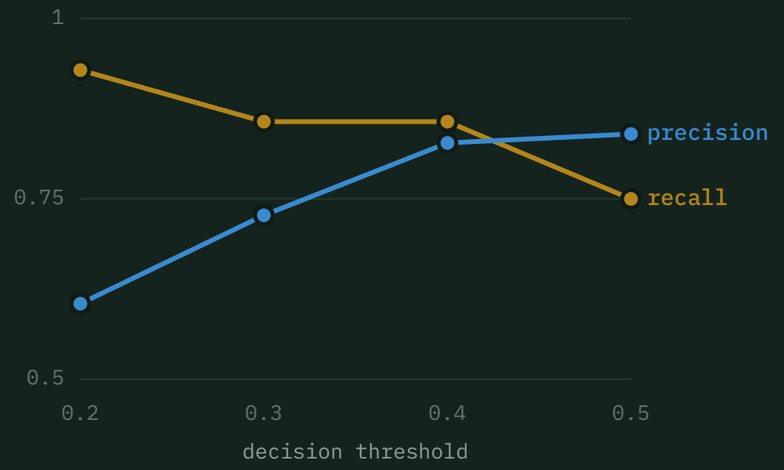
UNIT 2 · LINEAR MODELS FOR CLASSIFICATION

Seven classifiers, five folds, one target.

5-FOLD CV ROC-AUC (MEAN ± SD)



BAYESIAN LOGISTIC · THRESHOLD FOR EARLY WARNING



RBF-SVM grid search: C = 10, γ = 0.003 · CV AUC 0.954

PROJECT REVIEW 2

PBL-II

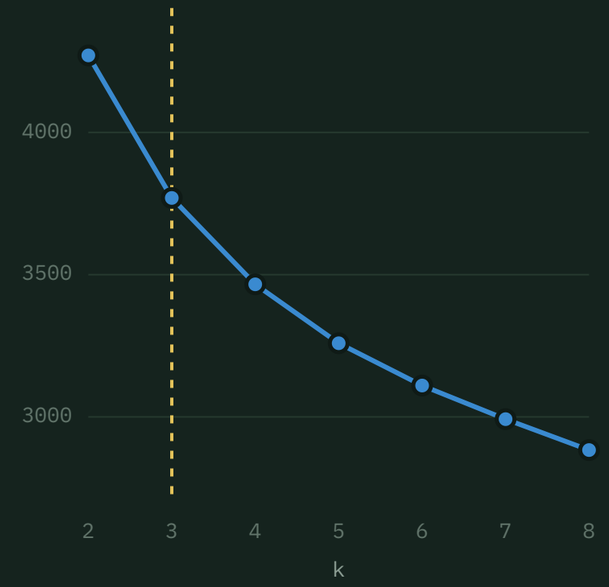
Units 3 & 4

K-Means | Gaussian mixtures · EM | Hierarchical | PCA · factor analysis | Markov chain | HMM · Baum–Welch · Viterbi | Kalman filter

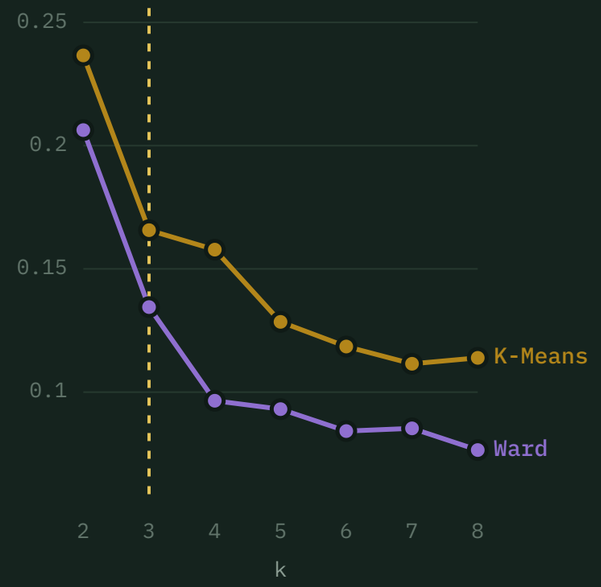
UNIT 3 · HOW MANY GROUPS?

Three criteria. BIC has the clearest answer: k = 3.

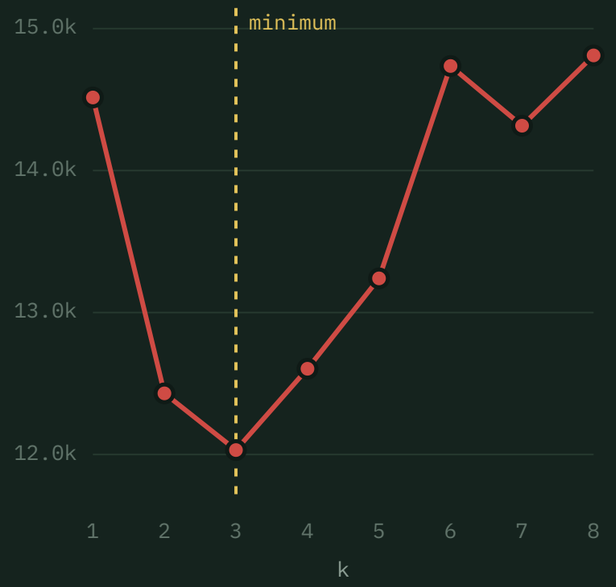
ELBOW · K-MEANS DISTORTION J



SILHOUETTE (HIGHER IS BETTER)

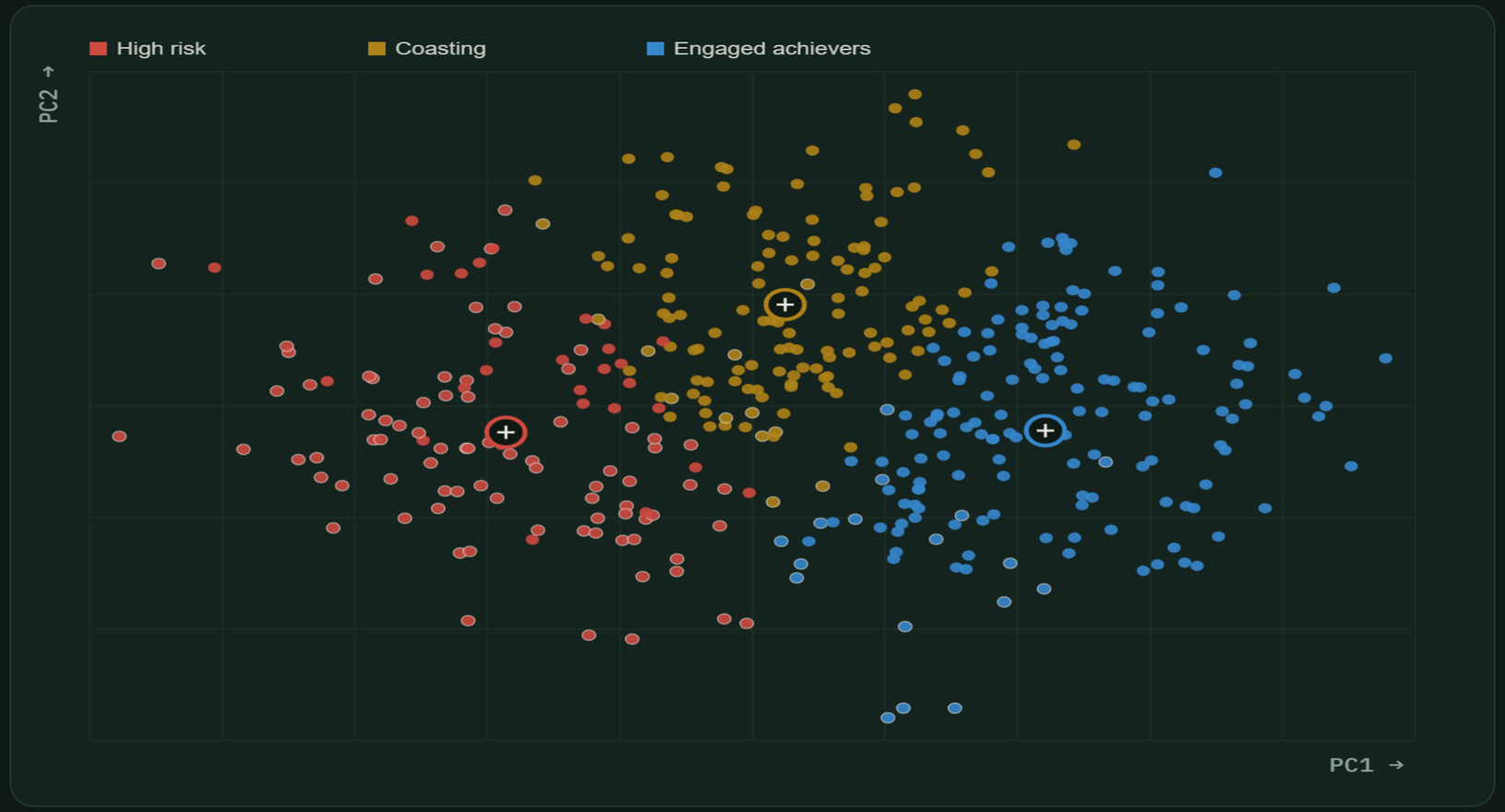


GMM BIC (LOWER IS BETTER)



UNIT 3 · K-MEANS FROM SCRATCH

Watch Lloyd's algorithm converge.



ITERATION

24 / 24

DISTORTION J

3776

CHECK

Scratch vs sklearn ARI **0.86**

↻ Replay

UNIT 3 · COMPARATIVE PERFORMANCE ANALYSIS

Eight clusterings, judged two ways.

METHOD	SILHOUETTE	DAVIES-B.	ARI RISK	ARI TRUE STATE	SIZES
K-Means (scratch)	0.162	1.85	0.220	0.174	111 / 160 / 129
K-Means (sklearn)	0.166	1.85	0.243	0.198	133 / 167 / 100
GMM / EM (scratch)	0.108	2.18	0.114	0.115	99 / 139 / 162
GMM (sklearn)	0.108	2.18	0.114	0.115	162 / 99 / 139
Hierarchical: Ward	0.135	2.04	0.225	0.162	186 / 104 / 110
Hierarchical: complete	0.131	2.15	0.195	0.108	138 / 72 / 190
Hierarchical: average	0.131	1.28	0.451	0.216	89 / 310 / 1
Hierarchical: single	0.138	0.66	0.005	0.000	398 / 1 / 1

SCRATCH EM VS SKLEARN

ARI 1.00

log-lik -4796.34 vs -4796.34

DISSIMILARITY (AVERAGE LINKAGE)

METRIC	SILH.	ARI
euclidean	0.131	0.451
manhattan	0.204	0.136
cosine	0.284	0.098

UNIT 3 · RESULT INTERPRETATION

No labels needed: three student personas.

High risk

111 students

74%

at risk this semester

low grades, low attendance, backlogs piling up

CGPA	5.9
attendance	73
internals	55
study h/day	1.5
late submissions	3.8
engagement	2.8
backlogs	1.6

Coasting

129 students

9%

at risk this semester

decent grades, little effort: the watch list

CGPA	7.6
attendance	81
internals	72
study h/day	1.5
late submissions	3.3
engagement	3.0
backlogs	0.1

Engaged achievers

160 students

11%

at risk this semester

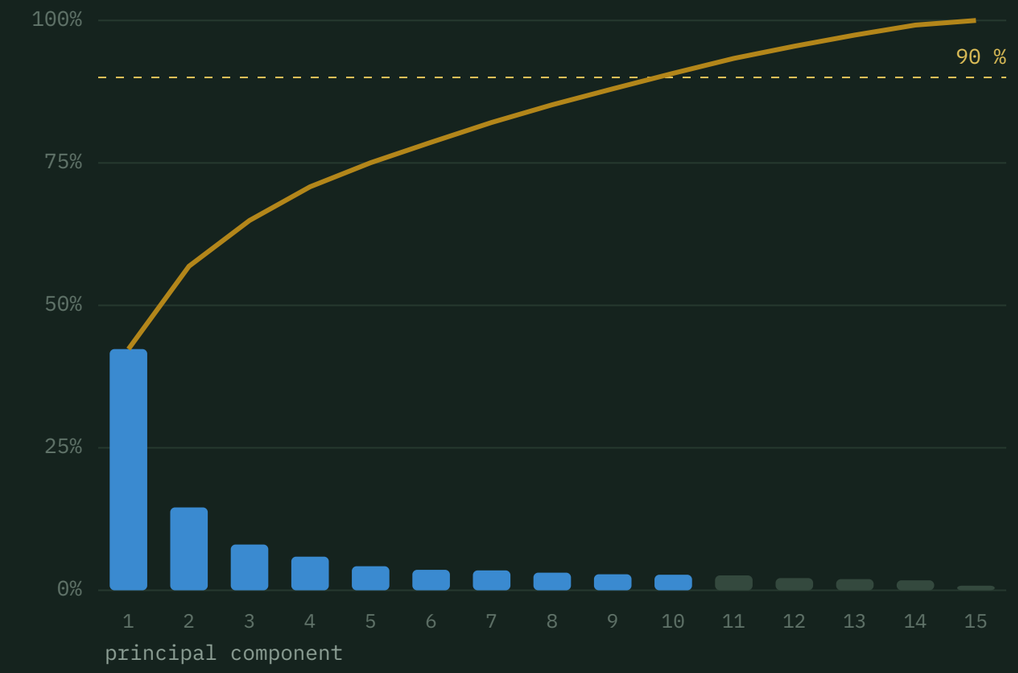
high effort and high grades

CGPA	8.1
attendance	89
internals	74
study h/day	2.9
late submissions	2.2
engagement	4.3
backlogs	0.1

UNIT 3 · LATENT DIMENSIONS & HIERARCHY

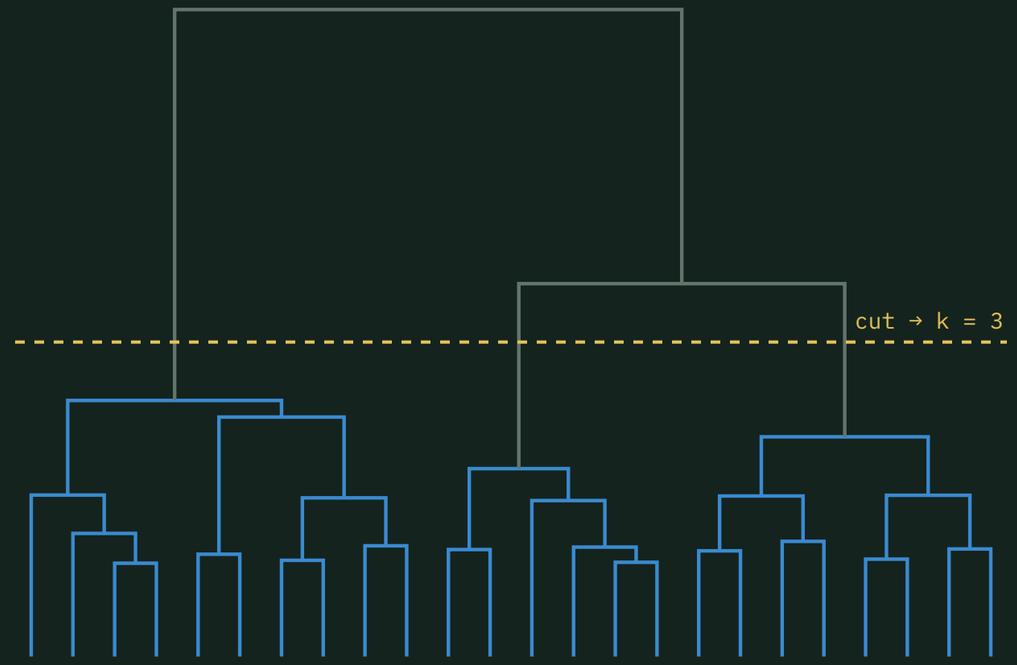
How many dimensions does risk really need?

SCREE · CUMULATIVE EXPLAINED VARIANCE



10 PCs for 90 % · 12 for 95 % · PC1 alone 42%

WARD DENDROGRAM · LAST 24 MERGES · CUT FOR K = 3



UNIT 3 · FACTOR ANALYSIS

28 questions, 4 hidden traits, recovered.

Cross-validated log-likelihood peaks at **5** factors. The 4-factor varimax solution:

FACTOR 1

Engagement

class participation	+0.75
late assignment frequency	-0.70
overall academic engagement	+0.69
online activity completion	+0.69
revision frequency	+0.68

FACTOR 2

Support

academic guidance frequency	+0.73
peer support	+0.72
faculty support	+0.71
academic help accessibility	+0.69
doubt resolution frequency	+0.43

FACTOR 3

Performance outlook

academic progress satisfaction	+0.77
academic performance concern	-0.55
programme completion certainty	+0.52
programme completion confidence	+0.51
academic workload difficulty	-0.50

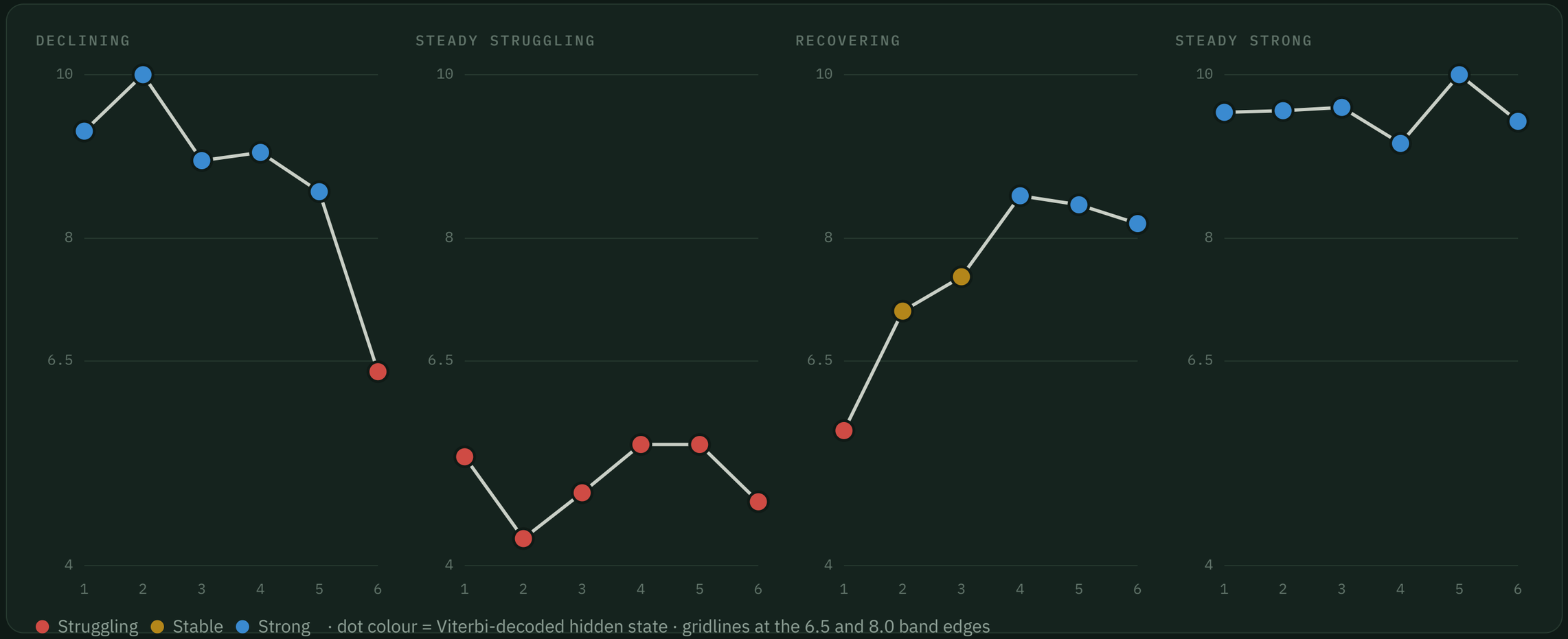
FACTOR 4

Motivation

academic motivation	+0.68
course interest	+0.63
career goal clarity	+0.60
self learning frequency	+0.53
additional resource usage	+0.47

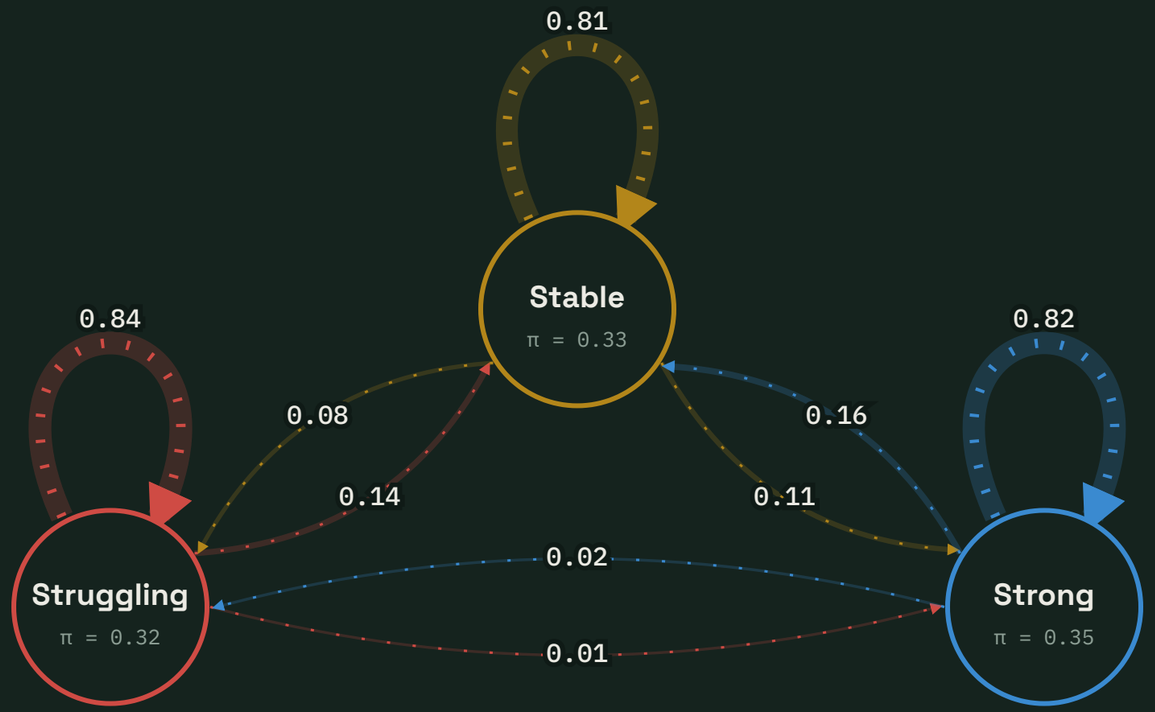
UNIT 4 · SEQUENTIAL DATA

A survey is a snapshot. A student is a trajectory.



UNIT 4 · BAUM–WELCH FROM SCRATCH

Struggling is sticky: 84% stay there next semester.



EMISSION MATRIX B

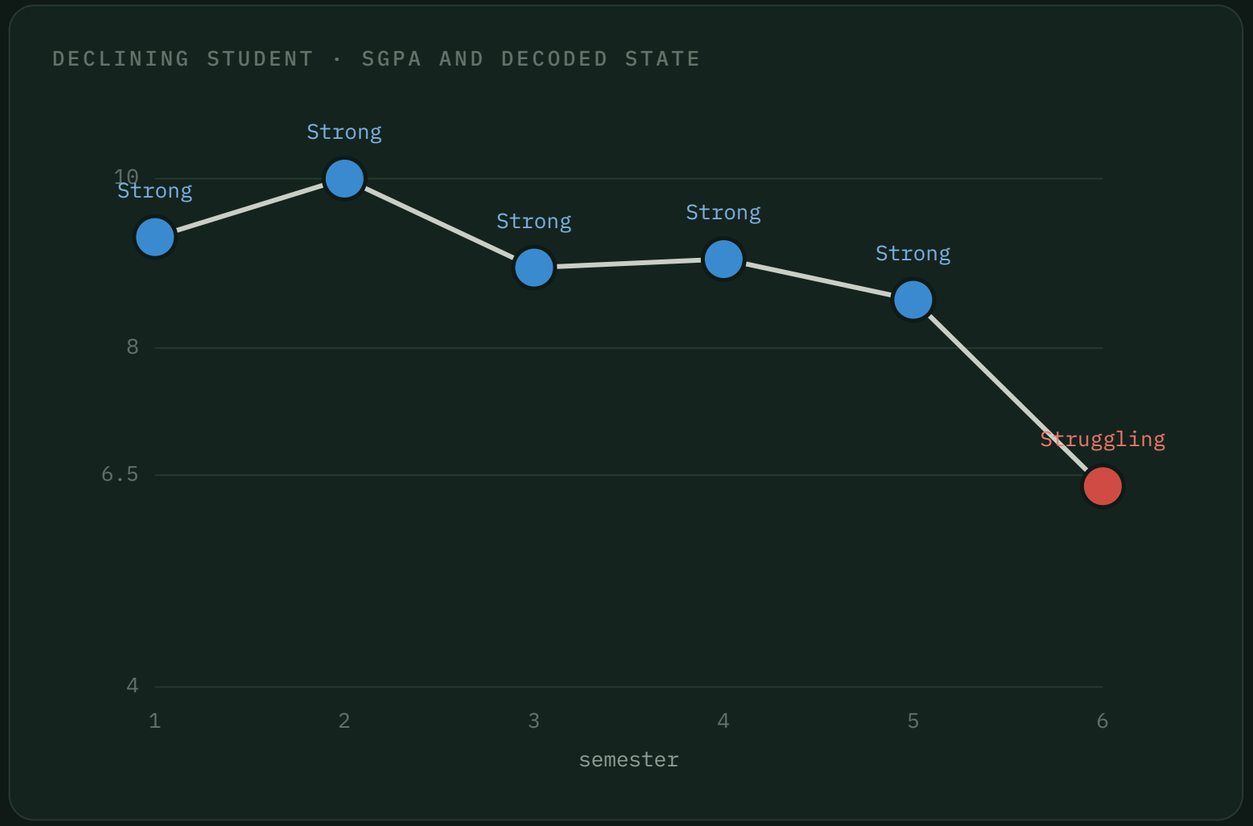
	POOR	AVERAGE	GOOD
Struggling	0.97	0.03	0.00
Stable	0.05	0.91	0.05
Strong	0.00	0.01	0.98

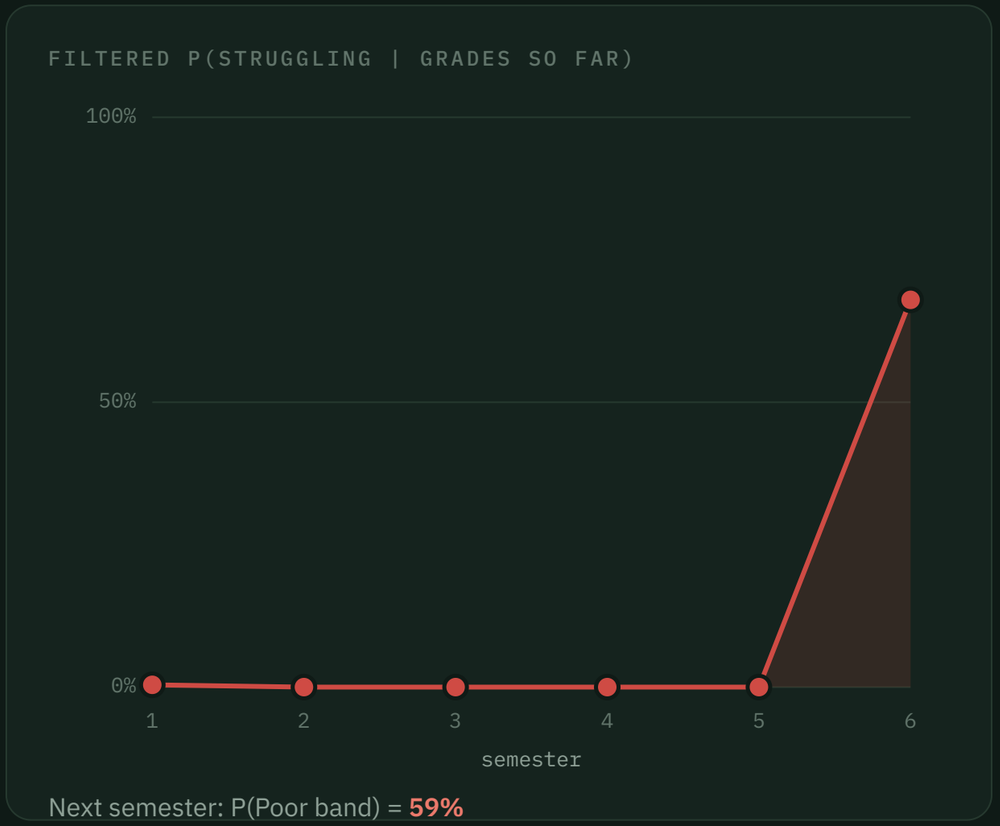
MODEL

π, A, B by **Baum–Welch** with scaling factors · 67 iterations · log-lik -1210.044

UNIT 4 · VITERBI & FORWARD FILTERING

Decoding the hidden state, one semester at a time.



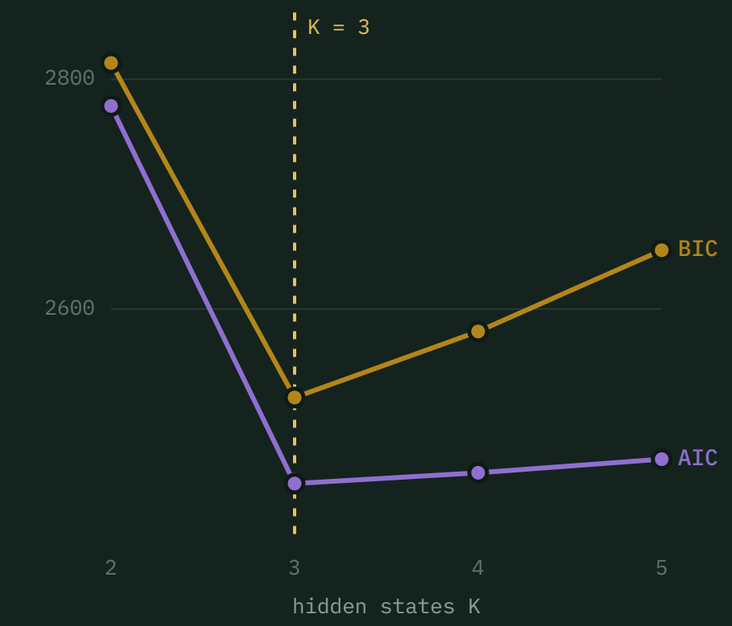


Next semester: P(Poor band) = **59%**

UNIT 4 · MODEL SELECTION & VERIFICATION

Right number of states. Right parameters. Right paths.

BIC VS NUMBER OF HIDDEN STATES K



LOG-LIKELIHOOD

-1210.044

scratch · hmmlearn: -1210.044

TRANSITION RECOVERY

0.063

max $|\hat{A} - A_{true}|$

VITERBI VS TRUE STATES

92.5%

of 1554 semesters

VITERBI AGREEMENT WITH HMMLEARN

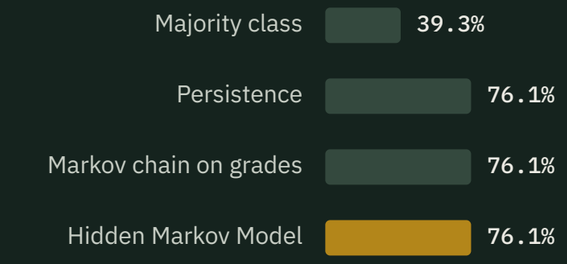
100%

identical decoded paths

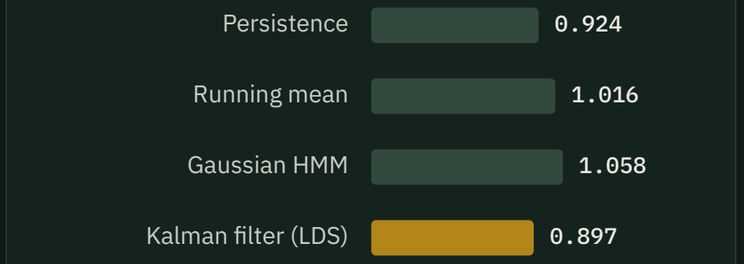
UNIT 4 · COMPARATIVE PERFORMANCE ANALYSIS

Forecasting next semester against honest baselines.

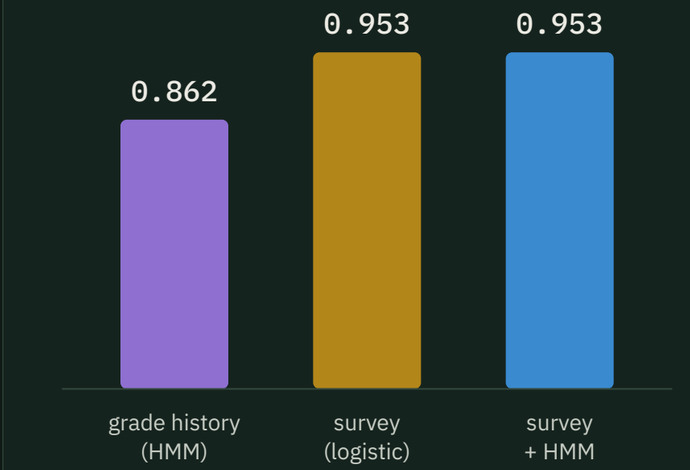
NEXT BAND · ACCURACY



NEXT SGPA · RMSE (LOWER IS BETTER)



THIS SEMESTER'S RISK · ROC-AUC



COMPARATIVE SUMMARY

Four units, one pipeline, every result checked.

UNIT	QUESTION	METHOD	RESULT
1	How much do two signals change the odds?	Bayes' rule	P(risk) 28% → 96%
2	Predict end-semester SGPA	Ridge	R^2 0.798
2	Classify at-risk students	Logistic Regression	CV AUC 0.953
3	How many groups?	GMM + BIC	k = 3
3	Groups without labels	K-Means (scratch)	74% · 9% · 11% at risk
3	How many dimensions?	PCA	10 PCs → 90 %
4	Hidden states over semesters	HMM (scratch)	K = 3 · Viterbi 92.5%
4	Next-semester SGPA	Kalman filter	RMSE 0.897

SELF-CHECKS

- ✓ schema 49 modelling columns
- ✓ no missing after cleaning
- ✓ ols normal equations match sklearn
- ✓ bayesian lr auc close to sklearn
- ✓ scratch gmm matches sklearn ari≥ 0.95
- ✓ scratch pca eigenvalues match sklearn
- ✓ scratch hmm loglik matches hmmlearn
- ✓ scratch hmm as good as hmmlearn fit

RESULT INTERPRETATION

What we learned, and what we can’t claim yet.

FINDINGS

- Internal marks (r = -0.67) and attendance (r = -0.59) are the strongest protective signals.
- Linear models already reach CV AUC ≈ 0.95; the Bayesian model adds a credible band to every prediction.
- Clustering finds a coasting group: decent grades, low effort, only 9% at risk today, but a natural watch list.
- Struggling is sticky: P(stay) = 0.84, the case for intervening early.
- Grade history alone predicts this semester's risk with AUC 0.86; the survey lifts it to 0.95.

LIMITS & ETHICS

- The cohort is **synthetic**; real responses so far are 15 high performers with no outcome label.
- Self-reported answers carry bias; correlation is not causation.
- Semester history comes from the generator; the form does not yet ask for per-semester SGPA.
- A risk score must only trigger an **offer of support**, never a label or a penalty.

LIVE DEMO

Thank you.
Questions?

Dashboard · localhost:8501

docker compose up

python run_all.py · 50 s on a laptop CPU

Next · PBL-III ensembles